Hybrid IS-VWAP Dynamic Algorithmic Trading via LQR *

Jackie Jianhong Shen
Electronic Trading Model Risk, Goldman Sachs
New York, New York, USA

June 10, 2017

Abstract

For optimal trade execution, static models are useful for pre-trade estimation or benchmarking, but real trading must be based on dynamic decision making or dynamic programming (DP). DP algorithms can adjust execution automatically to real-time market price waves as well as inventory positions, and thus are more responsive and realistic. Development of DP models however faces a few notorious obstacles including the curse of dimensionality, absence of closed-form solutions for general formulations, and difficulty in incorporating hard constraints. The DP approach based on linear-quadratic regulators (LQR) was first introduced to algorithmic trading by Hora [6]. In Dynamic Control Theory, LQR models allow closed-form solutions thanks to the Bellman equations and quadratic value functions. The resulted optimal policies are affine functions of the state variables. The current work improves the LQR model of Hora [6] in several important areas, including (a) the unconditional stability (or convexity), (b) retiring of the second-order small term of stagewise risk aversion, (c) quasi risk aversion achieved by time-varying delay costs, (d) incorporation of the bid-offer spreads, and (e) hybridizing the implementation shortfall (IS) algorithm with the VWAP approach to softly enforce the constraint of completion. The impact models and price dynamics still follow the earlier framework of Huberman and Stanzl [7]. The motivations behind every component of the improved model are elaborated, and the closed-form LQR solutions are derived (Theorem 1 and 2). Numerical simulation of the LQR trading paths is presented, which confirms several desirable properties of the hybrid IS-VWAP model.

Keywords: dynamic programming, LQR, IS, VWAP, slippage, spread, delay cost, impact cost, Bellman equation, value function, optimal policy, positive matrix, stability.

Attention: The current work is designed to be exclusively published in the Social Sciences Research Network (SSRN) preprint server. Any commercial or open journal is prohibited from publishing it without the prior endorsement from the author. As such, the paper in her current free form can enjoy some modern latex styles and enhance users' reading experiences, e.g., by creating book-style informative cover pages or colored text areas.

*Dr. Jackie Shen is an incoming member for Electronic Trading Model Risk at Goldman Sachs, New York, USA. Previously, he had been working for the equities Algorithmic Trading desks at both J.P. Morgan and Barclays, New York, USA. Email: jhshen@alum.mit.edu; URL: alum.mit.edu/www/jhshen

Electronic copy available at: https://ssrn.com/abstract=2984297
Electronic copy available at: https://ssrn.com/abstract=2984297

Contents

1 Introduction **4**

2 Basic Modeling Assumptions and Trading Settings **5**

3 Quest for a More Versatile LQR Execution Model **5**

4 The Improved IS Model via LQR **7**
4.1 Basic Market Settings 7
4.2 Market Impacts 8
4.3 State Variables and Transition 9
4.4 Construction of Stage Costs 10
4.5 Delay Costs 11
4.6 The LQR Model 12
4.7 The LQR Solution 12

5 The Hybrid IS-VWAP Model via LQR **16**

6 Computing and Simulation **17**

7 Disclaimers and Cautions **22**

Electronic copy available at: https://ssrn.com/abstract=2984297
Electronic copy available at: https://ssrn.com/abstract=2984297

The following terms are frequently used throughout the paper.

term	**meaning**
IS	implementation shortfall
VWAP	volume-weighted average price
PoV	participation of volume (rate)
COP	child order placement
DP	dynamic programming
LQR	linear quadratic regulator
H-ADV	average daily volume over a given horizon
NBBO	national best bid and offer (price)
ITM/OTM	in the money or out of the money
SEC	Securities and Exchange Commission (of the USA)
Bps	a basis point: 1 penny out of 100 dollars
"true" price	the mid point of the NBBO
horizon	the designated intraday window for execution
non-negative matrix	a symmetric real matrix with non-negative eigenvalues

The following symbols have been consistently used in the paper.

symbol	**meaning**	**controlled by**	**usage examples**
Δt	time bin size	house	$\sigma_{t,\Delta t}$
t	discrete or continuous time	model	J_t, V_t, x_t
S_t	raw security price	market	arrival price S_0
p_t	price slippage $p_t = (S_t - S_0)/S_0$	market	$p_0 = 0, p_T$
$\boldsymbol{x}_t$	state variable $\boldsymbol{x}_t = (q_t, p_t)'$	market, trading	$u_t^* = K_t \boldsymbol{x}_t + f_t$
$\sigma_{t,\Delta t}$	p_t's volatility over $[t, t+1)$	security, history	$\varepsilon_t = \sigma_{t,\Delta t} Z_t$
v_t	market volume over $[t, t+1)$	security, history	H-ADV$= v_0 + \ldots + v_{T-1}$
s_t	half spread relative to S_0	security, history	$s_t x_t / Q_0$
x_t	shares traded over $[t, t+1)$	model output	$x_0 + \ldots + x_{T-1}$
u_t	PoV rate over $[t, t+1)$	model output	$x_t = u_t v_t$
Q_t	outstanding shares at t	client, trading	$Q_{t+1} = Q_t - x_t$
q_t	outstanding % at t	client, trading	$Q_t = q_t Q_0$
a_t	intermediate quantity v_t/Q_0	model, scaling	$(a_t s_t) u_t$
μ_t	stagewise permanent impact	security, COP	see Eqn. (3)
η_t	stagewise instantaneous impact	security, COP	see Eqn. (4)
γ_t	cost on cumulative impact	house	see Eqn. (10)
β_t	stagewise delay cost	house	see Eqn. (15)
λ	overall delay cost/risk aversion	client, house	see Eqn. (16)

Electronic copy available at: https://ssrn.com/abstract=2984297
Electronic copy available at: https://ssrn.com/abstract=2984297

1 Introduction

In a complex and yet constantly evolving trading environment, algorithmic execution has to be *dynamic* in order to rationally swim through the dynamic waves of market prices or volumes, as well as one's own inventory positions. This has been discussed or demonstrated in a number of existing works, e.g., Bertsimas and Lo [4], and many others [1, 7, 6, 10, 5], just to cite a few.

The main tigers along the path of dynamic-programming (DP) based trading include the *curse of dimensionality* and the absence of closed-form solutions for most DP models. This has baffled any serious attempts of formulating self-contained and rigorous DP trading models that both allow tractable solutions and make actual economic sense. For example, there have been even no rigorous DP models with tractable solutions that could solidly implement the two basic client requests:

- **Completion**: to successfully acquire or liquidate a requested position Q_0 by a pre-determined ending time T, and
- **PoV Limit**: to trade within the participation-of-volume (PoV) limit range often specified by a savvy client, e.g., PoV $\leq 15.0\%$.

Let Q_t denote the residual position at time t that still needs to be filled. A naive scheme for *completion*, such as requesting on the last trading bin $[T-1, T)$ to trade $x_{T-1} = Q_{T-1}$, could be highly risky. It could temporarily deplete the market liquidity and stir up the flash-crash kind of market instabilities. Neither market regulators (e.g., SEC, FINRA, or CFTC in the US) nor clients would like to see this to happen.

Among all the DP trading models that allow explicit closed-form solutions and whose solutions also make actual financial sense, is the LQR approach first proposed by Hora [6]. Recall in *Dynamic Control Theory* that the *linear-quadratic regulator* (LQR) is a classic type of control problems and solutions under linear state transitions and additive quadratic cost functions [3].

Hora's LQR model is built upon the price dynamics initially proposed by Huberman and Stanzl [7], i.e., linear instantaneous price impacts settled into linear permanent impacts. Unlike Huberman and Stanzl [7] who targeted at the *global* mean-variance cost function (as by Almgren and Chriss [2]), Hora formulated the DP model using stagewise or localized mean-variance costs instead. The stagewise costs also incorporated a new quadratic term dubbed "the holding cost." In combination, Hora showed that under certain conditions the proposed LQR model can result in well-behaved solutions that actually react to price waves.

The current work attempts to improve Hora's model [6] in a number of important areas: (a) choice of working variables, (b) reformulation of holding costs, or what we call the "delay" cost, (c) incorporation of a new quadratic term to penalize market price slippages caused by cumulative permanent impacts, (d) inclusion of the bid-offer spread cost, and (e) hybridizing the improved IS algorithm with the popular VWAP algorithm. In combination, these improvements result in a novel hybrid IS-VWAP DP model

- that for each bin t directly outputs the optimal PoVs, instead of absolute shares;
- that softly enforces the completion constraint;
- that has an unconditionally stable LQR solution; and
- that trades in a style hybridizing both the IS and VWAP algorithms.

The rest of the paper has been organized as follows. The general model assumptions and trading settings are introduced in Section 2. In Section 3, we elaborate further on areas where Hora's model in [6] needs further improvements. We then develop in details the improved IS model based on the LQR framework in Section 4, and the hybrid IS-VWAP model in Section 5. Computational examples and simulation results are demonstrated in Section 6. The work is concluded with several disclaimers and cautions in Section 7.

Electronic copy available at: https://ssrn.com/abstract=2984297

2 Basic Modeling Assumptions and Trading Settings

The basic trading settings and key assumptions for the current work are listed as follows. In summary, the current work focuses on the optimal execution of a single name at the parent-slicing level, assuming an in-house *child order placement* (COP) scheme is already available.

(i) **(Single Name)** The current work focuses on *single-name* execution problems, preferably in the equity space where a single liquid stock or ETF is to be acquired or liquidated. The security must be liquid enough so that a continuous and positive intraday volume profile exists.

(ii) **(Running Example: Buy)** The running example throughout is a buy trade. For a sell trade, simply flip the signs of the residual positions Q_t's and trading shares x_t's or the PoVs u_t's. In actual implementation, one can introduce a new side variable, i.e., $side = +1$ for a buy trade and $side = -1$ for a sell trade.

(iii) **(Position Size)** We assume that the initial position Q_0 to be acquired or liquidated can be executed during an intraday window (i.e., "the horizon") without much risk of disrupting the market. This means that if the VWAP algorithm is to be adopted, the associated PoV rate over the execution horizon sits somewhere below 20% or 30%. PoV rates at the lower end enjoy more effective linear approximations, and thus allow us to adopt the linear impact models by Huberman and Stanzl [7]. This choice is the same as in Hora [6] and is fundamental for generating both linear state transitions and quadratic costs required by the LQR framework.

(iv) **(COP and SOR)** It is assumed that a *child order placement* (COP) scheme has been in place. The current work only deals with the parent slicing problem, and the output is a series of optimal trading policies that specify how many (child) shares x_t to trade on any incoming time bin $[t, t+1)$. The COP then manages the actual execution of the x_t shares over $[t, t+1)$, by a combination of sophisticated order placement strategies and a smart order routing (SOR) algorithm. The SOR is key for effectively accessing different liquidity venues. We refer to our earlier works in [10, 9] for related discussion on COP and SOR, and their potential impact on in-house model calibration.

(v) **(Child Bin)** We assume that an intraday execution horizon $[0, T]$ has been discretized to N equal-length child bins with duration Δt. For a liquid name, for instance, one can set $\Delta t = 30$ seconds or 1 minute.

To compress and simplify notations, unless a confusing scenario arises, the time variable t is used to represent both the continuous time and the associated discretized bin index. For example, $t+1$ means the next bin index from t, while $t+\Delta t$ means the same in the continuous setting. The same practice applies to the terminal time T, which could represent the actual continuous time or the discretized terminal index $N = T/\Delta t$.

Finally, we have listed on one of the cover pages both the terms and variables used throughout the paper frequently and consistently.

3 Quest for a More Versatile LQR Execution Model

In this section, some symbols follow the usage by Hora [6] for easier discussion. Their meaning can temporarily differ from the symbol list on one of the cover pages.

Electronic copy available at: https://ssrn.com/abstract=2984297

Working Variables

Most existing DP models, including Hora [6] and Huberman and Stanzl [7] for example, work with the trading shares x_t directly. But working with the PoV variable u_t is more natural and beneficial for the following reasons.

First, the market impact caused by shares x_t is always relative to a security's liquidity level. The total market volume v_t is a good proxy for gauging market liquidity (though it could be deceptive during disruptive trading sessions as cautioned in SEC's investigation reports). Then market impacts can be better formulated on the PoV variable:

$$u_t = \frac{x_t}{v_t}, \qquad \text{for any time bin } t.$$

When the PoVs are moderate, both the permanent and instantaneous market impacts can be reasonably approximated by linear functions of u_t.

Another advantage of working with the PoV variable u_t is that model parameters can remain invariant for different trades of a same security. Ideally model parameters should depend only on the market or security information, but not on specific trade requests such as the position size Q_0.

The PoV control u_t also allows the solutions to integrate seamlessly with any in-house volume predictor $\hat{V}_t$, which is an indispensable component in any heuristic implementation of a DP model. The trading position at t is then given by $x_t = u_t \hat{V}_t$, and clearly reacts to the volume $\hat{V}_t$.

Finally, in Section 5 we shall demonstrate that the PoV variable u_t serves as a bridge between IS and VWAP that results in a more effective hybrid IS-VWAP model.

Flat Delay Costs and Quasi Risk Aversion

Let Q_t denote the residual position still outstanding at time t. Then

$$Q_{t+1} = Q_t - x_t, \qquad t = 0, 1, \ldots.$$

In [6], Hora first introduced a penalty term into the stage cost j_t:

$$j_t = \rho Q_t^2 + \text{other terms}.$$

In the current work, we shall call it the delay cost.

With a possible intraday discounting factor $\beta \in (0, 1]$ as introduced by Hora [6], which is applied to j_t as a powered weight β^t, one sees that the delay costs remain flat (i.e., ρ) or decreasing (i.e., $\beta^t \rho$) even as the execution gets closer to the designated ending time T.

This is a bit counter-intuitive since most clients do care about successful completion by a designated time, e.g., $T =$ 3:30pm. Ideally, the weights for the delay costs should head up instead of remaining flat or decreasing, when an execution approaches its ending time T.

In the current work, we shall discuss how to design monotonically increasing weights for the delay costs. In particular, we show how such a design can create a proxy for risk aversion that previously can only be implemented by static pre-trade models [2, 7].

Necessity of Stagewise Risk Aversion

Inspired by the global mean-variance framework of Huberman and Stanzl [7] as well as Almgren and Chriss [2], Hora introduced the stagewise mean-variance costs (ref. Eqn. (5) in Hora [6]):

$$j_t = E\left[\Delta_t \frac{x_t}{Q_0}\right] + \frac{\lambda}{2}\text{var}\left(\Delta_t \frac{x_t}{Q_0}\right) + \rho Q_t^2,$$

Electronic copy available at: https://ssrn.com/abstract=2984297

where Δ_t denotes the execution price shortfall (i.e., execution price above the arrival price).

For the well-behavedness of the model, Hora argues in [6] that it requires $Q_0 \gg 1$ (ref. Eqn. (8) in Hora [6]). Then the first term for the mean is of order $O(1/Q_0)$, but the second term of variance is only of order $O(1/Q_0^2)$. This forces the above stagewise risk aversion parameter λ to depend on the trading position Q_0 in order to materialize the variance effect (e.g., $\lambda = O(Q_0)$).

This is a bit undesirable since ideally model parameters should depend only on the market or the trading settings (e.g., choice of a horizon), but not on any specific trade request.

For this reason, the local variance term will be retired in the current model. Instead, we will revamp the delay cost term so that it can play a similar role of risk aversion.

Lack of Positivity and Unconditional Stability

In the settings of Hora's model [6], the stagewise costs are quadratic but not positive definite (ref. Eqn. (7) and the matrix form between Eqn. (9) and (10) in Hora [6]). The three eigenvalues of the matrix are:

$$\lambda_0 = \rho, \qquad \lambda_\pm = \frac{\theta}{2} \pm \frac{1}{2}\sqrt{\theta^2 + Q_0^{-2}},$$

where $\rho, \theta > 0$ are model parameters in [6]. In particular, $\lambda_- < 0$.

As a result, there is no mechanism to guarantee the positivity of the Riccati matrices (i.e., v_t's in Eqn. (11) of Hora [6]). Then the inversion of $\theta + \beta B' v_{t+1} B$ in Eqn. (12) of [6] can be unstable. An inequality condition was proposed in Eqn. (8) of [6], which however involves an unknown random variable Δ_T at the terminal time. Thus the well-behavedness cannot hold unconditionally.

The solution we propose here is to introduce an extra quadratic cost into the stagewise cost function j_t. It penalizes the market price slippages caused by the cumulative permanent impacts, and makes the model unconditionally stable.

It is now ready to design the improved IS model based on the LQR approach.

4 The Improved IS Model via LQR

Throughout we assume that the main task is to execute Q_0 shares of a given single name during a designated intraday horizon. For convenience, the horizon is normalized to a time interval $[0, T]$, and T is called the terminal time. Notice that in general $t = 0$ does not have to be the start of the open trading session, e.g., 9:30am for the equity market in the USA. In fact, a savvy client would usually prefer to stay away from the volatile opening minutes.

As in Section 2, it is assumed that a child window length Δt has been chosen (e.g., 2 minutes), and the horizon is discretized to time bins indexed by $n = 0, 1, \ldots, N$ with $N\Delta t = T$. As explained earlier, to compress notation usage, we shall use the same symbol t for both the continuous time and the associated discrete bin index n $(= t/\Delta t)$. Thus for example, we will denote the bins also by $0, \ldots, t, \ldots, T$, and $t + 1$ refers to the next bin following the t-bin.

4.1 Basic Market Settings

The current work does not address specifically stochastic volumes. Instead, we assume that the average daily volume for the horizon (H-ADV) is M shares, and that the volume profile is given by

$$v_0, \ldots, v_t, \ldots, v_{T-1}, \qquad \text{with } v_t \text{ shares on } [t, t+1). \tag{1}$$

In particular, one has $M = v_0 + \ldots + v_{T-1}$.

Electronic copy available at: https://ssrn.com/abstract=2984297

Let S_t denote the *true* price of the security at time t, and S_0 the arrival price. In reality, S_t can be identified with the mid price of the National Best Bid and Offer (NBBO) prices. Let

$$p_t = \frac{S_t - S_0}{S_0}$$

denote the price slippage per dollar. For intraday trading, p_t is assumed to follow a Brownian motion:

$$p_{t+1} = p_t + \sigma_{t,\Delta t} Z_t, \tag{2}$$

where $\sigma_{t,\Delta t}$ refers to the volatility over the single bin $[t, t+\Delta t)$, and Z_t a canonical Gaussian with zero mean and unit variance. Z_t's are assumed to be independent for $t = 0, 1, \ldots, T-1$. If the intraday volatility turns out to be flat, one can drop the bin notation by writing $\sigma_{\Delta t}$. For example, if σ_{day} denotes a single-day volatility and a single day is normalized to one unit of time, one has: $\sigma_{\Delta t} = \sigma_{day}\sqrt{\Delta t}$. For execution houses with high data processing capacities, one could study and calibrate more realistic intraday volatility profiles.

4.2 Market Impacts

Following Hora [6], we also employ the *linear* impact models of Huberman and Stanzl [7]. These models are simple and applicable for trades with low PoVs, as discussed earlier. They result in linear state transitions and quadratic cost functions required by the LQR framework.

Let x_t denote the shares to be traded over an incoming bin $[t, t+1)$. Then

$$u_t = \frac{x_t}{v_t} \qquad \text{is the PoV rate over the bin.}$$

Both impact models are expressed via this PoV variable.

Under the trading activity u_t, the dynamics of the price slippage in Eqn. (2) is now updated to:

$$p_{t+1} = p_t + \mu_t u_t + \sigma_{t,\Delta t} Z_t, \tag{3}$$

where μ_t is the coefficient for permanent impact. In actual implementation (such as in the numerical simulation in Section 6), μ_t should be benchmarked by the bin volatility $\sigma_{t,\Delta t}$, and hence rewritten by: $\mu_t = \mu_t^0 \sigma_{t,\Delta t}$ with a normalized coefficient μ_t^0. Furthermore, the time subscript t can be dropped if it is flat.

The permanent impact over the t-bin only affects all trades after $t+1$. It is the "instantaneous" impact that directly costs the current trade x_t or u_t over $[t, t+1)$, which is modeled by:

$$\tilde{p}_{t+0.5} = p_t + s_t + \eta_t u_t + \omega_t. \tag{4}$$

The components are explained as follows.

(i) $\tilde{p}_{t+0.5}$ is the average execution slippage (with respect to the arrival price S_0) for all the x_t shares executed over the t-bin $[t, t+1)$. The notation 0.5 in the subscript emphasizes that it is not a point-in-time quantity but associated with a duration.

(ii) s_t denotes the half spread in relative to the arrival price S_0. It has been added to the slippage p_t of the true price S_t (or the mid quote of the NBBO). In the current work, the spread s_t is assumed to be deterministic and can be approximated from historic intraday profiles.

This slippage component was neglected in both works of [7] and [6]. For even liquid securities traded near 10 dollars, for example, the "Sub-Penny Rule" of the SEC (*Securities and Exchange Commission* of the USA) requires quotes be no finer than a penny, which results in a spread about 10 Bps. Recall that a basis point (Bps) is one penny out of 100 dollars.

Electronic copy available at: https://ssrn.com/abstract=2984297

(iii) η_t is the instantaneous impact coefficient. At the micro level, it reflects the premium cost for liquidity taking when the trading event progressively reveals itself to the market and before the limit order book (LOB) restores itself at $t+1$. For liquid securities, in general one should have

$$\eta_t \geq \mu_t, \text{ i.e., instantaneous impact is greater than the final settled permanent impact.}$$

This is because market participants will inject new liquidity into the temporarily depleted high quote levels. Often this is called market resilience. (The actual transition mechanisms are discussed in earlier works such as [8] or [9]). η_t should be benchmarked by the bin volatility $\sigma_{t,\Delta t}$. In the numerical simulations later, we have adopted $\eta_t = \eta_t^0 \sigma_{t,\Delta t}$ with $\eta_t^0 = 1.0$.

(iv) Finally, the stochastic component ω_t reflects the general behavior of market participants when the market fluctuates around its equilibrium. In both [6] and [7], ω_t was assumed to be a Gaussian with zero mean and independent of the random component Z_t associated with the true price in Eqn. (3).

We argue that ω_t and Z_t should be highly correlated, instead. For example, when the COP program executes trading at a constant rate over the child bin $[t, t+1)$ and when the POV is low and thus negligible, the theory of Brownian bridge should result in:

$$\omega_t = 0.5\sigma_{t,\Delta t} Z_t,$$

meaning that it actually correlates with Z_t 100%! In the current work, we only require ω_t to have a zero mean. In particular,

$$E[\tilde{p}_{t+0.5} \mid p_t] = p_t + s_t + \eta_t u_t. \tag{5}$$

4.3 State Variables and Transition

Let Q_t denote the outstanding shares that still need to be filled at time t. Define the residual rate:

$$q_t = \frac{Q_t}{Q_0}, \qquad t = 0, 1, \ldots, T.$$

Notice that the initial value is always $q_0 = 100\%$, and ideally one wishes for $q_T = 0$ for promised completion.

In this work, the **state** variable $\boldsymbol{x}_t$ consists of the price slippage rate p_t and the inventory residual rate q_t:

$$\boldsymbol{x}_t := (q_t, p_t)',$$

where the superscript means vector or matrix transposition. The first component reflects the inventory position and the second the market environment.

The **control** variable is the PoV rate $u_t = x_t/v_t$ defined in the preceding subsection, where x_t is the number of shares to be executed over the t-bin $[t, t+1)$ and v_t the total market shares over the same bin. In this work, market volumes are assumed to be deterministic and established via historic intraday volume profiles.

As in the preceding subsection, the transition of the price slippage is given by:

$$p_{t+1} = p_t + \mu_t u_t + \varepsilon_t, \qquad \text{with} \quad \varepsilon_t = \sigma_{t,\Delta t} Z_t. \tag{6}$$

For the inventory residual rate q_t, one has

$$q_{t+1} = \frac{Q_{t+1}}{Q_0} = \frac{Q_t - x_t}{Q_0} = q_t - a_t u_t, \qquad \text{with} \quad a_t = v_t/Q_0. \tag{7}$$

Electronic copy available at: https://ssrn.com/abstract=2984297

In combination, the complete state transition is expressed by

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - a_t \boldsymbol{v}_t u_t + (0, \varepsilon_t)', \qquad \text{with} \quad \boldsymbol{v}_t = (1, -\mu_t/a_t)'. \tag{8}$$

It is a linear transition with an additive stochastic component of zero mean. The introduction of the quantity a_t will simplify the expressions for the final solutions.

4.4 Construction of Stage Costs

We seek a stagewise cost function in the form of:

$$J_0 = \sum_{t=0}^{T-1} j_t + j_T,$$

where the last term j_T is the terminal cost. In general, at any time t, the forward cost is

$$J_t = \sum_{\tau=t}^{T-1} j_\tau + j_T,$$

Let $x_t = u_t v_t$ denote the number of shares to be executed over the t-bin, with a PoV rate u_t and market volume v_t. Then conditioning on the state at t, the expected execution slippage of all these shares is, thanks to Eqn. (5),

$$E\left[\tilde{p}_{t+0.5}\left(\frac{x_t}{Q_0}\right) \mid p_t, q_t\right] = a_t u_t E\left[\tilde{p}_{t+0.5}\right] = (a_t \eta_t) u_t^2 + a_t p_t u_t + (a_t s_t) u_t. \tag{9}$$

This term is the main backbone of the stagewise cost j_t. Unlike the model by Hora [6] which is based upon the localized mean-variance framework, we will not incorporate the associated conditional variance into j_t as explained in the preceding section.

Instead, we introduce an extra quadratic cost to the price slippage p_t:

$$(a_t \gamma_t) p_t^2, \qquad \text{with} \quad \gamma_t \eta_t > \frac{1}{4}. \tag{10}$$

The inequality condition on the weight parameter γ_t is to ensure that the two terms in Eqn. (9) and (10) can be integrated into a *positive and definite* quadratic form:

$$(a_t \eta_t) u_t^2 + a_t p_t u_t + (a_t s_t) u_t + (a_t \gamma_t) p_t^2 = a_t (p_t, u_t) \begin{pmatrix} \gamma_t & \frac{1}{2} \\ \frac{1}{2} & \eta_t \end{pmatrix} \begin{pmatrix} p_t \\ u_t \end{pmatrix} + (a_t s_t) u_t. \tag{11}$$

Define $m = 4\gamma_t \eta_t - 1$. Then

$$\gamma_t = \frac{1+m}{4\eta_t}, \qquad \text{with} \quad m > 0.$$

This can be used to design γ_t with any positive free parameter m. Notice that $m/4$ is exactly the determinant of the quadratic matrix in Eqn. (11).

This new cost term penalizes the *cumulative* permanent impact, as explained below. From the dynamics of the price slippage in Eqn. (3), one has

$$p_t = p_0 + \sum_{\tau=0}^{t-1} \sigma_{\tau,\Delta t} Z_\tau + \sum_{\tau=0}^{t-1} \mu_\tau u_\tau,$$

Electronic copy available at: https://ssrn.com/abstract=2984297

where p_0 actually vanishes since $p_t = (S_t - S_0)/S_0$. Since the second term of Brownian motion reflects market behavior and is independent of the execution plan, to penalize p_t via p_t^2 thus amounts to penalizing the last term of cumulative permanent impacts.

Similar to Hora [6], we wrap up model design by adding the third component of delay cost:

$$\beta_t q_t^2, \qquad \text{with} \quad \beta_t > 0. \tag{12}$$

In Hora [6], it was called the *holding* cost and had a constant weight. In the next subsection, we elaborate on the necessity and actual design of a time-varying weight β_t.

In combination, the stage cost j_t is defined by

$$j_t = \lambda \beta_t q_t^2 + a_t(p_t, u_t) \begin{pmatrix} \gamma_t & \frac{1}{2} \\ \frac{1}{2} & \eta_t \end{pmatrix} \begin{pmatrix} p_t \\ u_t \end{pmatrix} + (a_t s_t) u_t, \tag{13}$$

for any $t < T$. At the end of the execution horizon $t = T$, the terminal cost is given by

$$j_T = \lambda \beta_T q_T^2, \tag{14}$$

since upon termination there will be no further execution u_T beyond T, and $a_T = v_T/Q_0$ is no longer applicable.

The new constant parameter $\lambda > 0$ balances the relative weights between the delay cost and the slippage and spread costs. Its role will be further elucidated next.

4.5 Delay Costs

We now discuss how to properly design the delay cost weights β_t's, which must be time varying.

In the extreme case if one sets $\beta_T = +\infty$ at the terminal, completion is then enforced, i.e.,

$$q_T = 0, \quad \text{or equivalently, } Q_T = 0.$$

However, this may disrupt the market since it demands on the last execution bin $[T-1, T)$:

$$x_{T-1} = Q_{T-1} - Q_T = Q_{T-1},$$

i.e., liquidating whatever positions still outstanding by $T-1$. It could deplete the market liquidity if the outstanding position Q_{T-1} is substantive.

In reality, to *softly* enforce the completion condition, we shall design the β_t's so that they are monotonically increasing and penalize more any delayed executions. The objective is to achieve some degree of *quasi risk aversion* as in the traditional mean-variance algorithms [7, 2, 9].

This line of reasoning inspires the following approach to achieving quasi risk aversion. At time t, the variance of the price slippage p_t grows to

$$\sigma_t^2 = \sum_{\tau=0}^{t-1} \sigma_{\tau,\Delta t}^2$$

Notice that it grows linearly with t if intraday volatilities are assumed to be flat. In general, this variance series increase monotonically. At any time t, σ_t^2 represents the uncertainty level of the price slippage that the residual position Q_t is exposed to, at least *on paper* (i.e., without considering the impact it may self inflict).

Furthermore, to have an extra degree of freedom to internally distribute delay costs, we introduce a new parameter $\rho \in (0, 1]$ and define

$$\beta_t = \rho^{T-t} \sigma_t^2, \qquad t = 0, 1, \ldots, T. \tag{15}$$

Electronic copy available at: https://ssrn.com/abstract=2984297

For ρ's strictly less than 1.0, the algorithm can be more reactive to price slippages instead of being compelled to trade more aggressively in the beginning (i.e., the typical front-loading behavior under traditional risk aversion). This is because the delay penalties become relatively negligible in the beginning phase of an execution. For example, for $\rho = 0.95$ and $T = 120$ minutes, one has at $t = 0$: $\beta_0 = \rho^{T-0} = 0.0021$. The simulation results in Section 6 confirm this desirable behavior.

4.6 The LQR Model

To summarize, we observe the following behavior about the stagewise cost j_t in Eqn. (13).

- **Impact Cost.** Both the second and third terms favor *slow* trading in order to reduce costs associated with instantaneous and permanent impact costs, as well as the spread cost.
- **Market Risk.** The first term for delay cost, however, encourages *fast* trading, as any delay in execution raises cost.

Thus like the more traditional mean-variance framework [2, 7, 9], the current model can potentially achieve a balance between the impact cost and the market risk, even without the localized variance term of Hora [6].

It then becomes clear that the constant parameter λ plays a role very similar to the risk aversion parameter in traditional mean-variance models. We have coined the term of "quasi" risk aversion for this special role.

To conclude the design, the global objective function for the new model is given by:

$$J_\tau = \sum_{t=\tau}^{T} j_t = \sum_{t=\tau}^{T} \left(\lambda \beta_t q_t^2 + a_t(p_t, u_t) \begin{pmatrix} \gamma_t & \frac{1}{2} \\ \frac{1}{2} & \eta_t \end{pmatrix} \begin{pmatrix} p_t \\ u_t \end{pmatrix} + (a_t s_t) u_t \right). \tag{16}$$

A bigger λ penalizes more the delay cost, and hence achieves a higher level of risk aversion.

Next, we seek optimal execution policies that are purely state driven:

$$u_t^* = \phi_t(q_t, p_t), \quad t = 0, 1, \ldots, T-1,$$

so that

$$\left(u_0^*, \ldots, u_{T-1}^*\right) = \text{argmin}_{(u_0,\ldots,u_{T-1})} E[J_0\left(u_0, \ldots, u_{T-1}\right) \mid q_0, p_0].$$

The expectation is over the sample space of market price waves: $(Z_0, \ldots, Z_{T-1})$, as in Eqn. (6).

4.7 The LQR Solution

Define the value function V_t by:

$$V_t(q_t, p_t) = \min_{(u_t,\ldots,u_{T-1})} E[J_t\left(u_t, \ldots, u_{T-1}\right) \mid q_t, p_t].$$

Then one has to solve the Bellman equations:

$$V_t(q_t, p_t) = \min_{u_t} j_t(u_t \mid q_t, p_t) + E\left[V_{t+1}(q_{t+1}, p_{t+1}) \mid q_t, p_t, u_t\right]. \tag{17}$$

Electronic copy available at: https://ssrn.com/abstract=2984297

The Solution at $T-1$ and Interpretation

To get a direct sense about the solution, we first manually work out the solution at $T-1$.

The value function at the terminal is given by:

$$V_T = j_T = \lambda \beta_T q_T^2.$$

Set $t = T-1$, and recall the state transition in Eqn. (7). Then the optimal policy $u_t^* = u_{T-1}^*$ should minimize:

$$\lambda \beta_t q_t^2 + a_t(p_t, u_t) \begin{pmatrix} \gamma_t & \frac{1}{2} \\ \frac{1}{2} & \eta_t \end{pmatrix} \begin{pmatrix} p_t \\ u_t \end{pmatrix} + (a_t s_t) u_t + \lambda \beta_T (q_t - a_t u_t)^2.$$

The solution to this quadratic optimization at $t = T-1$ is:

$$\begin{aligned} u_{T-1}^* &= A_{T-1} q_{T-1} + B_{T-1}(p_{T-1} + s_{T-1}), \qquad \text{with} \\ A_{T-1} &= \frac{\lambda \beta_T}{\eta_{T-1} + \lambda a_{T-1} \beta_T}, \qquad \text{and} \quad B_{T-1} = \frac{-0.5}{\eta_{T-1} + \lambda a_{T-1} \beta_T}. \end{aligned} \tag{18}$$

In particular, when the penalty for the delay cost at terminal $\beta_T \to +\infty$, one has

$$u_{T-1}^* \to \frac{1}{a_{T-1}} q_{T-1} + 0(p_{T-1} + s_{T-1}) = \frac{Q_0}{V_{T-1}} \frac{Q_{T-1}}{Q_0} = \frac{Q_{T-1}}{V_{T-1}},$$

meaning that one trades all the Q_{T-1} shares still outstanding at $T-1$: $x_{T-1}^* = u_{T-1}^* V_{T-1} = Q_{T-1}$.

Furthermore, one notices the following desirable properties from the explicit solution.

- The optimal policy is *positively* proportional to the outstanding position, i.e., $A_{T-1} > 0$. As a result, it trades more if there are more residual positions still outstanding.
- It is *negatively* proportional to the market price slippage (for the running example of a buy trade), i.e., $B_{T-1} < 0$.

  Let us look into the meaning in a realistic trading context. Since the true market price slippage p_t is associated with the mid quote of the NBBO and s_t is the half spread, $p_t + s_t$ therefore corresponds to the price slippage of the national best offer (NBO). As our running example is a buy trade, the NBO represents the execution price for a buying *market* order. The negativity of B_{T-1} means to buy less when the NBO shoots up with a positive slippage, and more when the opposite occurs.

  Such behavior is clearly in line with the heuristic but popular street practice of trading aggressively while in the money (ITM) (and passively while out of the money (OTM)).

An important observation is that the denominator involved in the solution:

$$g_{T-1} = \eta_{T-1} + \lambda a_{T-1} \beta_T,$$

is strictly positive. This makes the solution well-posed and stable. We will establish the same property for the general case coming next.

Electronic copy available at: https://ssrn.com/abstract=2984297

The Complete LQR Solution

Since the objective function J is not in the standard or popular form of LQR, we give the full details on the Bellman solution process. The standard formulation of LQR in the literature always assumes a *separable* quadratic form for the stage costs, with no cross interaction between the state and control variables [3]. In our model, however, the cross term $a_t p_t u_t$ that couples the state variable p_t and control variable u_t represents the slippage costs and hence plays a key role.

First we recall the state transition established earlier in Eqn. (8) (with $a_t = v_t/Q_0$ as defined in Eqn. (7)):

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - a_t u_t \boldsymbol{v}_t + \begin{pmatrix} 0 \\ \varepsilon_t \end{pmatrix}, \qquad \text{with} \quad \boldsymbol{x}_t = \begin{pmatrix} q_t \\ p_t \end{pmatrix} \qquad \text{and} \quad \boldsymbol{v}_t = \begin{pmatrix} 1 \\ -\mu_t/a_t \end{pmatrix}. \tag{19}$$

Also the stochastic component is given by $\varepsilon_t = \sigma_{t,\Delta} Z_t$, with Z_t's being i.i.d. canonical Gaussians.

Theorem 1 *At each time t, there exist a non-negative 2 by 2 matrix P_t, a 2 by 1 vector $\boldsymbol{b}_t$, a scalar e_t, such that the value function V_t is given by:*

$$V_t(q_t, p_t) = \boldsymbol{x}_t' P_t \boldsymbol{x} + \boldsymbol{b}_t' \boldsymbol{x}_t + e_t = (q_t, p_t) P_t \begin{pmatrix} q_t \\ p_t \end{pmatrix} + \boldsymbol{b}_t' \begin{pmatrix} q_t \\ p_t \end{pmatrix} + e_t. \tag{20}$$

The optimal policy is given by the affine form:

$$\begin{aligned} u_t^* &= \phi_t(\boldsymbol{x}_t) = K_t \boldsymbol{x}_t + f_t, \qquad \textit{with} \\ K_t &= \frac{1}{g_t}(\boldsymbol{v}_t' P_{t+1} - (0, 0.5)), \quad \textit{(a single-row matrix)} \quad \textit{and} \\ f_t &= \frac{0.5}{g_t}(\boldsymbol{b}_{t+1}' \boldsymbol{v}_t - s_t), \end{aligned} \tag{21}$$

where s_t denotes the half spread as before, and

$$g_t = \eta_t + a_t \boldsymbol{v}_t' P_{t+1} \boldsymbol{v}_t.$$

Furthermore, the structure of the value functions cascades backwards as follows:

$$\begin{aligned} P_t &= P_{t+1} + \Delta_t - a_t g_t K_t' K_t, \\ \boldsymbol{b}_t &= \boldsymbol{b}_{t+1} - 2 a_t g_t f_t K_t', \qquad \textit{and} \\ e_t &= e_{t+1} + c_{t+1}^{(2,2)} \sigma_{t,\Delta t}^2 - a_t g_t f_t^2, \end{aligned} \tag{22}$$

where $c_{t+1}^{(2,2)}$ denotes the (2,2)-element of P_{t+1}, and $\Delta_t = \mathrm{diag}(\lambda\beta_t, a_t\gamma_t)$. The terminal data are:

$$P_T = \mathrm{diag}(\lambda\beta_T, 0), \quad \boldsymbol{b}_T = (0, 0)', \qquad \textit{and} \quad e_T = 0. \tag{23}$$

Proof. First, one defines

$$h_t(u_t \mid \boldsymbol{x}_t) = E_{\varepsilon_t}[V_{t+1}(\boldsymbol{x}_{t+1}) \mid \boldsymbol{x}_t, u_t].$$

Then the Bellman equation in Eqn. (17) becomes:

$$V_t = \min_{u_t} j_t(u_t \mid \boldsymbol{x}_t) + h_t(u_t \mid \boldsymbol{x}_t).$$

Electronic copy available at: https://ssrn.com/abstract=2984297

Assuming V_{t+1} is indeed in the form of Eqn. (20) and noticing the state transition in Eqn. (19), one has

$$h_t(u_t \mid \boldsymbol{x}_t) = (\boldsymbol{x}_t - a_t u_t \boldsymbol{v}_t)' P_{t+1}(\boldsymbol{x}_t - a_t u_t \boldsymbol{v}_t) + \boldsymbol{b}'_{t+1}(\boldsymbol{x}_t - a_t u_t \boldsymbol{v}_t) + \left(e_{t+1} + c_{t+1}^{(2,2)} \sigma_{t,\Delta t}^2\right). \tag{24}$$

On the other hand, the stage cost in Eqn. (16) can be rewritten as a quadratic function of u_t:

$$j_t(u_t \mid \boldsymbol{x}_t) = \boldsymbol{x}'_t \Delta_t \boldsymbol{x}_t + a_t(p_t + s_t)u_t + a_t \eta_t u_t^2. \tag{25}$$

Rearrange all the terms in $j_t(u_t \mid \boldsymbol{x}_t) + h_t(u_t \mid \boldsymbol{x}_t)$ to a canonical quadratic function of u_t:

$$\begin{aligned}
j_t + h_t &= C - 2Bu_t + Au_t^2, \qquad \text{with} \\
C &= \boldsymbol{x}'_t(P_{t+1} + \Delta_t)\boldsymbol{x}_t + \boldsymbol{b}'_{t+1}\boldsymbol{x}_t + \left(e_{t+1} + c_{t+1}^{(2,2)} \sigma_{t,\Delta t}^2\right), \\
B &= a_t \left(\boldsymbol{v}'_t P_{t+1} \boldsymbol{x}_t + 0.5 \boldsymbol{b}'_{t+1} \boldsymbol{v}_t - 0.5(p_t + s_t)\right), \qquad \text{and} \\
A &= a_t \left(\eta_t + a_t \boldsymbol{v}'_t P_{t+1} \boldsymbol{v}_t\right) = a_t g_t.
\end{aligned}$$

Here g_t is as introduced in the theorem.

As a result, the optimal policy is given by $u_t^* = B/A$, which can be easily expressed as an affine function of the state variable $\boldsymbol{x}_t$:

$$u_t^* = K_t \boldsymbol{x}_t + f_t, \qquad \text{where } K_t \text{ and } f_t \text{ are as given in Eqn. (21).}$$

Furthermore, the value function is given by:

$$\begin{aligned}
V_t(\boldsymbol{x}_t) &= j_t(u_t^* \mid \boldsymbol{x}_t) + h_t(u_t^* \mid \boldsymbol{x}_t) \\
&= C - A\left(u_t^*\right)^2 = C - A(K_t \boldsymbol{x}_t + f_t)^2 \\
&= \boldsymbol{x}'_t(P_{t+1} + \Delta_t)\boldsymbol{x}_t + \boldsymbol{b}'_{t+1}\boldsymbol{x}_t + \left(e_{t+1} + c_{t+1}^{(2,2)} \sigma_{t,\Delta t}^2\right) - a_t g_t (\boldsymbol{x}'_t K'_t + f_t)(K_t \boldsymbol{x}_t + f_t)
\end{aligned}$$

Rearranging it into the quadratic form of the state variable $\boldsymbol{x}_t$ gives the cascading formulae in Eqn. (22).

Finally, let us prove the non-negativity of the P_t's. This clearly holds for P_T. For a general $t < T$, recall that

$$V_t = \min J_t = \min \sum_{\tau=t}^{T} j_\tau,$$

and by Eqn. (16),

$$j_\tau = \lambda \beta_\tau q_\tau^2 + a_\tau (p_\tau, u_\tau) \begin{pmatrix} \gamma_\tau & 0.5 \\ 0.5 & \eta_\tau \end{pmatrix} \begin{pmatrix} p_\tau \\ u_\tau \end{pmatrix} + (a_\tau s_\tau) u_\tau.$$

By design, the (strict) positivity of the middle term implies that there exists a time-independent positive constant $d > 0$, such that for any $\tau = 0, \ldots, T-1$,

$$(p_\tau, u_\tau) \begin{pmatrix} \gamma_\tau & 0.5 \\ 0.5 & \eta_\tau \end{pmatrix} \begin{pmatrix} p_\tau \\ u_\tau \end{pmatrix} \geq d(p_\tau^2 + u_\tau^2).$$

Consequently,

$$j_\tau \geq (a_\tau d) u_\tau^2 + (a_\tau s_\tau) u_\tau \geq -\frac{1}{4d} a_\tau s_\tau^2.$$

Electronic copy available at: https://ssrn.com/abstract=2984297

This also holds at $\tau = T$ since $j_T \geq 0$. As a result, for any value of the state variable $\boldsymbol{x}_t$,

$$V_t = V_t(\boldsymbol{x}_t) \geq -\frac{1}{4d} \sum_{\tau=t}^{T} a_\tau s_\tau^2.$$

Since we have just established recursively that V_t is given by the quadratic form of $\boldsymbol{x}_t$ in Eqn. (20), this lower bound reveals that P_t has to be non-negative. This completes the proof. □

In particular, the common denominator g_t for the optimal policy is strictly positive:

$$g_t = \eta_t + a_t \boldsymbol{v}_t' P_{t+1} \boldsymbol{v}_t \geq \eta_t > 0,$$

where η_t is the instantaneous impact coefficient introduced in Eqn. (4). This ensures that both K_t and f_t, and hence also u_t^* are well defined and unconditionally stable.

5 The Hybrid IS-VWAP Model via LQR

Assume that one is given a task of buying Q_0 shares for a security and an intraday horizon $[0, T]$ endorsed by both the client and the execution house. Let $M = v_0 + v_1 + \ldots + v_{T-1}$ denote the total market shares over the horizon (e.g., the H-ADV). Then the PoV rate for a VWAP algorithm is:

$$\text{PoV} = \frac{Q_0}{M}.$$

A pure VWAP algorithm would require: $\quad u_t \equiv \text{PoV}, \quad t = 0, 1, \ldots, T-1.$

If we incorporate this VWAP strategy softly into the implementation shortfall (IS) model in the preceding section by, for $t = 0, 1, \ldots, T-1$,

$$\begin{aligned} \tilde{j}_t &= j_t + a_t \theta (u_t - \text{PoV})^2 \\ &= \lambda \beta_t q_t^2 + a_t (p_t, u_t) \begin{pmatrix} \gamma_t & \frac{1}{2} \\ \frac{1}{2} & \eta_t \end{pmatrix} \begin{pmatrix} p_t \\ u_t \end{pmatrix} + (a_t s_t) u_t + a_t \theta (u_t - \text{PoV})^2, \end{aligned} \tag{26}$$

with a VWAP weighting parameter $\theta > 0$, one ends up with a hybrid IS-VWAP model. Here a_t in this new term is for order normalization, as coordinated with the second and third terms above.

> When the weight θ is sufficiently substantive, this hybrid model trades in a neighborhood of the VWAP algorithm, meaning that the optimal PoV policies must stay in a vicinity of the VWAP PoV. In particular, this can softly enforce the completion constraint since the VWAP PoV is designed to achieve so. This contrasts with most DP trading models that often lose touch with the completion requirement, except for ones that are too heavily risk averse and hence miss price improvement opportunities at the later stages of execution.

As the VWAP weight $\theta \to +\infty$, clearly this hybrid model recovers the VWAP algorithm. The general solution to this new hybrid IS-VWAP model is given below in Theorem 2, which is very similar to Theorem 1.

The major updates from Theorem 1 have been highlighted in red. The proof follows exactly the same steps as for Theorem 1 and has been omitted.

The solution clearly reveals the following properties.

- (**IS**) When $\theta \to +0$, the solution converges to the IS model in Theorem 1.

Electronic copy available at: https://ssrn.com/abstract=2984297

- (**VWAP**) When $\theta \to +\infty$ and hence $g_t \simeq \theta \to \infty$, it can be easily seen that:

$$K_t \to 0, \qquad \text{and} \quad f_t \to \text{PoV}.$$

  Indeed the optimal policy converges to the VWAP algorithm: $u_t^* \equiv \text{PoV}$.

- (**Hybrid**) In between, it is a hybrid model between IS and VWAP. A client or the execution house can tune the VWAP weight θ to achieve a desirable inclination. But in order to softly honor the completion constraint, as discussed earlier, one should work with a sufficiently heavy weight θ. This becomes especially important when the quasi risk aversion parameter λ is not strong enough.

Theorem 2 *At each time t, there exist a non-negative 2 by 2 matrix P_t, a 2 by 1 vector $\boldsymbol{b}_t$, a scalar e_t, such that the value function V_t is given by:*

$$V_t(q_t, p_t) = \boldsymbol{x}_t' P_t \boldsymbol{x} + \boldsymbol{b}_t' \boldsymbol{x}_t + e_t = (q_t, p_t) P_t \begin{pmatrix} q_t \\ p_t \end{pmatrix} + \boldsymbol{b}_t' \begin{pmatrix} q_t \\ p_t \end{pmatrix} + e_t. \tag{27}$$

The optimal policy is given by the affine form:

$$\begin{aligned} u_t^* &= \phi_t(\boldsymbol{x}_t) = K_t \boldsymbol{x}_t + f_t, \qquad \textit{with} \\ K_t &= \frac{1}{g_t}(\boldsymbol{v}_t' P_{t+1} - (0, 0.5)), \quad \textit{(a single-row matrix)} \quad \textit{and} \\ f_t &= \frac{0.5}{g_t}(\boldsymbol{b}_{t+1}' \boldsymbol{v}_t - s_t + 2\theta \text{PoV}), \end{aligned} \tag{28}$$

where s_t denotes the half spread as before, and

$$g_t = \eta_t + a_t \boldsymbol{v}_t' P_{t+1} \boldsymbol{v}_t + \theta.$$

Furthermore, the structure of the value functions cascades backwards as follows:

$$\begin{aligned} P_t &= P_{t+1} + \Delta_t - a_t g_t K_t' K_t, \\ \boldsymbol{b}_t &= \boldsymbol{b}_{t+1} - 2 a_t g_t f_t K_t', \qquad \textit{and} \\ e_t &= e_{t+1} + c_{t+1}^{(2,2)} \sigma_{t,\Delta t}^2 - a_t g_t f_t^2 + \theta a_t \text{PoV}^2, \end{aligned} \tag{29}$$

where $c_{t+1}^{(2,2)}$ denotes the (2,2)-element of P_{t+1}, and $\Delta_t = \text{diag}(\lambda \beta_t, a_t \gamma_t)$. The terminal data are:

$$P_T = \text{diag}(\lambda \beta_T, 0), \quad \boldsymbol{b}_T = (0, 0)', \qquad \textit{and} \quad e_T = 0. \tag{30}$$

The development in this section also demonstrates the power of using the PoV variable u_t as the control variable, instead of the absolute shares x_t as in works such as [7, 6]. It reaches out seamlessly to popular algorithms such as the PoV or the WAP.

6 Computing and Simulation

In this section we simulate several generic trading paths output from the proposed hybrid IS-VWAP model. The assumed market and trading settings are as shown in Table 1 below.

Electronic copy available at: https://ssrn.com/abstract=2984297

Table 1: Market and trading settings for the simulation

variable	**simulation value**
day in minutes	390 (9:30am to 4:00pm, typical for the US equity market)
single name symbol	ABC_XYZ (hypothetical)
ADV	3,000,000 (shares)
arrival price S_0	\$20.00
raw half spread	half penny (\$0.005)
daily volatility	1.00% (with roughly annual volatility 16.0%)
trade side	buy
trade position Q_0	100,000 (shares)
horizon in minutes	120 (11:00am to 1:00pm)
time bin size Δt	1 minute

By assuming *flat* intraday volatilities, we obtain the bin volatility:

$$\sigma_{t,\Delta t} = \sigma_{\Delta t} = \sigma_{day} \cdot \sqrt{\frac{1}{390}} = 5.1 \text{ Bps}. \tag{31}$$

And at any time t, the volatility of the price slippage p_t is given by $\sigma_t = \sigma_{\Delta t} \cdot \sqrt{t}$. Similarly, assuming a flat intraday volume distribution, one obtains the horizon ADV and the VWAP PoV:

$$\text{H-ADV} = ADV \cdot \frac{T}{390} = 923,077, \qquad \text{and} \quad \text{PoV} = \frac{Q_0}{\text{H-ADV}} = 10.8\%. \tag{32}$$

Listed in Table 2 are model parameters used in the simulation. Following the above assumption on flat intraday volatilities, the impact parameters are assumed to be time independent. Also as discussed earlier, $\mu_t < \eta_t$ due to market resilience.

Table 2: Default parameters for market impacts or trading style in the simulation

parameter symbol	**meaning or control**	**simulated value**
η_t	stagewise instantaneous impact	$\eta_t = \eta^0 \cdot \sigma_{t,\Delta t}$ with $\eta^0 = 1.0$
μ_t	stagewise permanent impact	$\mu_t = \mu^0 \cdot \sigma_{t,\Delta t}$ with $\mu^0 = 0.8$
γ_t and m	cost of cumulative impact	$\gamma_t = \frac{1+m}{4\eta_t}$ with $m = 1.0$
β_t and ρ	stagewise delay cost	$\beta_t = \rho^{T-t} \cdot \sigma_t^2$ with $\rho = 1.0$
λ	overall delay cost (quasi risk aversion)	$\lambda = 1000.0$
θ	VWAP weight	$\theta = 1.0$

In addition, a moderate U-shaped volume profile has been assumed over the 2-hour trading horizon $[0, T]$ (see the blue volume profiles in the upper panels of the following three figures). Its effect on the simulation is minor though it does provide a more realistic flavor.

Figure 1, 2, and 3 illustrate three generic trading paths under different choices of the trading or model parameters. In general, they confirm the desirable effects induced by the major trading parameters, including quasi risk aversion by the delay cost parameter λ.

When the quasi risk aversion is not too strong, the traditional front-loading behavior is weakened, and the DP algorithm becomes more sensitive and reactive to price waves. The captions of these three figures elaborate more on these details.

Electronic copy available at: https://ssrn.com/abstract=2984297

On the Soft Implementation of the Completion Constraint

In terms of softly enforcing the completion constraint, both λ and θ play a key role. A stronger λ enhances risk aversion and front loading, and hence in general helps complete the execution even before the ending time T. However, such early completion may come at the potential cost of missing later opportunities of price improvement.

Thus a better way to honor completion is to employ both a moderately strong λ and a moderately strong θ. The resulting moderate front loading helps execute a major portion in the beginning phase. At the later stages of execution, the moderately strong θ ensures that the hybrid model trades in a neighborhood of the VWAP model. This means that the optimal PoV rate u_t^* will stay in the vicinity of the VWAP PoV rate (i.e., 10.8% for these examples), and hence the algorithm virtually trades above some minimum PoV rate. In the three examples in Figure 1, 2, and 3, for instance, the algorithms all trade above the PoV of 5% or 6%. This effectively prohibits idling and promotes timely completion during the later stages of execution.

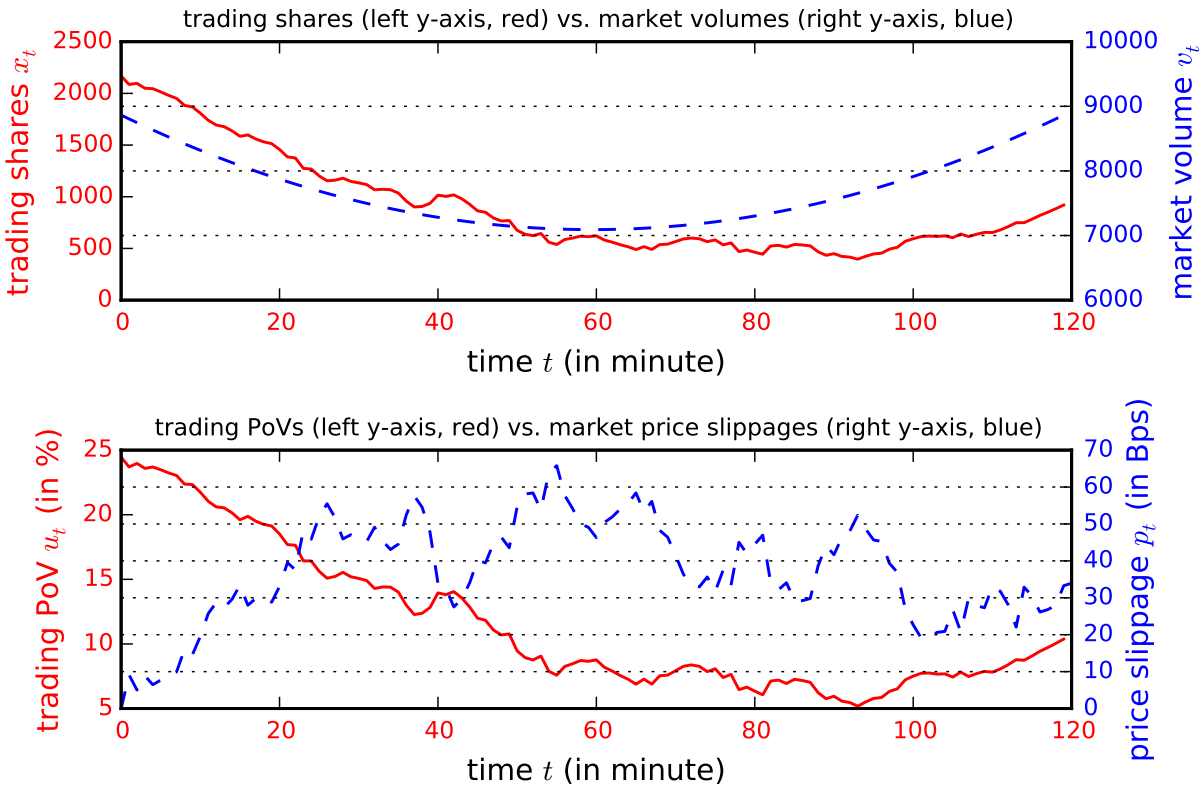

Figure 1: A single trading path by the hybrid IS-VWAP algorithm, simulated with the default parameters in Table 2. As explained in Section 4, the parameter λ for the delay cost serves as a proxy for traditional risk aversion. This results in the front-loading behavior, i.e., the red lines for trading shares x_t (in the upper panel) and trading PoVs u_t (in the lower panel).

Electronic copy available at: https://ssrn.com/abstract=2984297

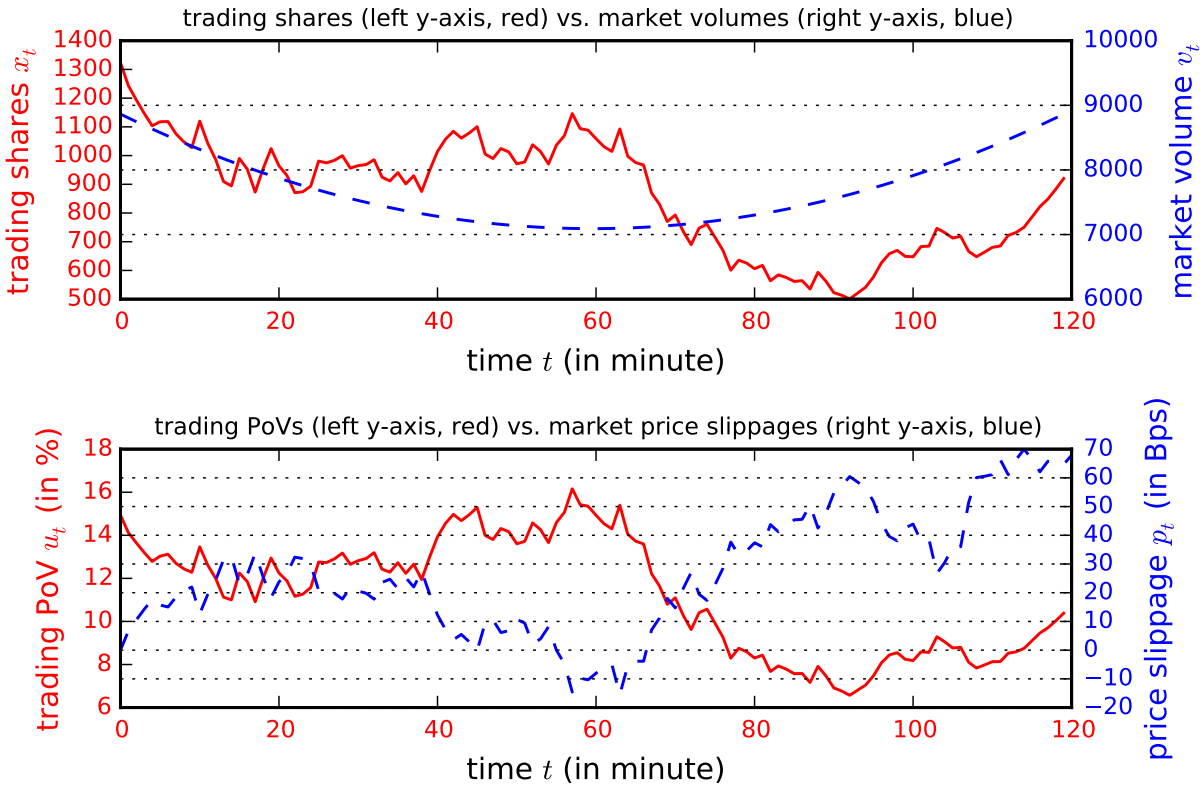

Figure 2: A single trading path by the hybrid IS-VWAP algorithm with the same default parameters as for Figure 1, except for ρ which is updated from 1.0 to 0.98. This reduces the strength of the quasi risk aversion λ as the combination $\lambda \cdot \rho^{T-t}$ is now less. As observed from the plotting, the front loading behavior is weakened, and the algorithm becomes more responsive to price waves. It trades more aggressively when the price slippages drop, and slows down when they start to hike.

Electronic copy available at: https://ssrn.com/abstract=2984297

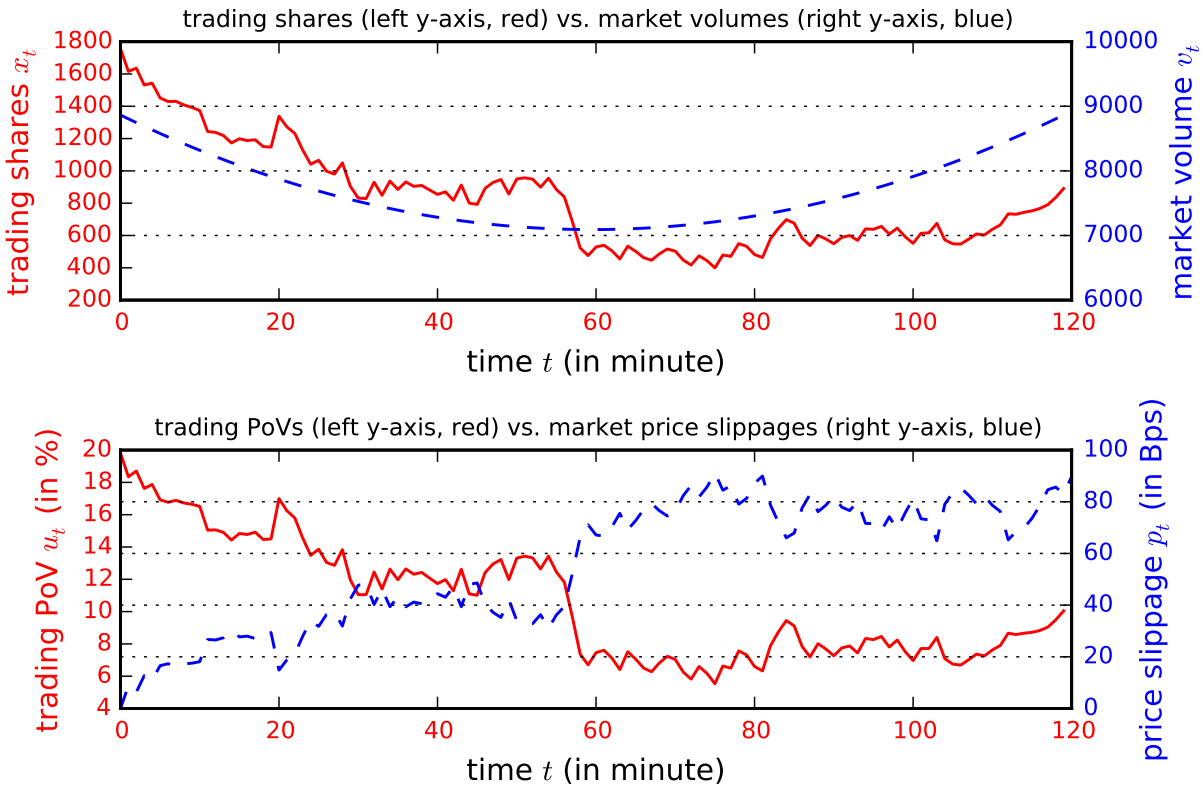

Figure 3: A single trading path by the hybrid IS-VWAP algorithm with the same parameters as for Figure 2, except for the delay cost λ and the VWAP weight θ, which are updated from $\lambda = 1000.0$ to 1500.0 and from $\theta = 1.0$ to 0.5. The smaller θ value reduces the VWAP effect (with PoV $= 10.8\%$ as in Eqn. (32)) and allows a broader range of PoV variations compared with Figure 2. On the other hand, the increment of the delay cost λ is to softly enforce front loading and hence the completion constraint. **Notice how the model properly reacts to the sudden price hike near** $t = 60$.

Electronic copy available at: https://ssrn.com/abstract=2984297

7 Disclaimers and Cautions

We conclude the current work with a few necessary disclaimers and cautions.

(1) The proposed trading model has not been the internal or external product of any execution houses where the author had worked. Any potential industrial conflict should be promptly directed to the attention of the author.

(2) Due to the proprietary nature of the industry and the resulting scarcity of real trading data to the public, the current academic work does not dive deeper into the data aspects. Interested execution houses are thus encouraged to further tailor the current framework to the available trade or market databases, e.g., to use dynamically projected volumes, or to calibrate the impact cost parameters via in-house trading records.

(3) While the outputs (i.e., the optimal PoV policies u_t^*'s) can be heuristically applicable to even stochastic volumes, the current model itself is not built upon stochastic volumes nor stochastic volatilities. Interested readers are refer to works such as Almgren [1] for related explorations.

(4) As the current work focuses mainly on the closed-form solutions of the LQR approach, any actual implementation must take necessary post-processing steps to make the solutions more practical and operationally safer. For example, one may have to floor or cap the optimal PoV policies u_t^*'s using a minimum or maximum PoV rate. Similarly, the DP nature of the algorithm in general cannot guarantee completion before the horizon ends, though many of our simulation experiments have shown that completion is generally achieved when the quasi risk aversion parameter λ is strong enough, or when λ and θ are both moderately strong.

(5) Execution houses who are interested in the current work and plan to implement it in their systems should be aware of any other securities or operational risks that are not specifically addressed in the current work, including for examples, IPO's (initial public offerings) without much trading histories, complex market opens or closes, trading halts, or any unexpected disruptive market events, etc.

Acknowledgments

Jackie Shen is grateful to Merrell Hora for the initial introduction to the field of dynamic algorithmic trading and for encouraging him to experiment with its multiple facets, while working together at the Barclays' equities electronic trading desk.

Jackie Shen is also very grateful to all the colleagues previously at the electronic trading desks of J.P. Morgan and Barclays, in particular many of our hard-working IT and RISK colleagues whose names hardly appear in the headlines, for their daily professional assistance as well as generous personal support. Reliable and healthy electronic trading would be impossible without solid IT implementations or effective risk controls.

Electronic copy available at: https://ssrn.com/abstract=2984297

References

[1] R. Almgren. Optimal trading with stochastic liquidity and volatility. *SIAM J. Financial Math.*, 3:163–181, 2012.

[2] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *J. Risk*, 3:5–39, 2000.

[3] D. P. Bertsekas. *Dynamic Programming and Optimal Control.* Athena Scientific, 2007.

[4] D. Bertsimas and A. W. Lo. Optimal control of execution costs. *J. Financial Markets*, 1(1):1–50, 1998.

[5] B. Bouchard, N.-M. Dang, and C.-A. Lehalle. Optimal control of trading algorithms: a general impulse control approach. *SIAM J. Finan. Math.*, 2(1):404–438, 2011.

[6] Merell Hora. Tactical liquidity trading and intraday volume. *Preprint*, pages 1–28, 2006.

[7] G. Huberman and W. Stanzl. Optimal liquidity trading. *Yale School of Management Working Papers*, YSM 165, 2001.

[8] A. Obizhaeva and J. Wang. Optimal trading strategy and supply/demand dynamics. *J. Financial Markets*, 16(1):1–32, 2013.

[9] J. Shen. A pre-trade algorithmic trading model under given volume measures and generic price dynamics. *Applied Math. Res. eXpress,* Oxford Univ. Press, 2015(1):64–98, 2015.

[10] J. Shen and Y. Yu. Styled dynamic algorithmic trading and the MV-MVP style. *Social Sci. Res. Network (SSRN) Preprint*, (2507002):1–18, 2014.

Electronic copy available at: https://ssrn.com/abstract=2984297